

Lingbo Tong

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RESEARCH INTERESTS	Natural Language Processing, Deep Learning, Structural Equation Modeling, Longitudinal Data Analysis, Computational Social Science.	
ACADEMIC APPOINTMENTS	University of Wisconsin-Madison , Madison, WI Quantitative Methods, Department of Educational Psychology Assistant Professor Anna Julia Cooper Research Fellow	Incoming, 2026 2025–2026
EDUCATION	University of Notre Dame , Notre Dame, IN Joint Ph.D. in Quantitative Psychology and Computer Science and Engineering <i>Advisors: Prof. Zhiyong (Johnny) Zhang and Prof. Meng Jiang</i> M.S. in Computer Science and Engineering Sichuan University , Chengdu, China B.Eng. in Computer Science and Technology	2025 2024 2020
PUBLICATIONS	<i>Under Review / In Preparation</i> Tong, L. , & Zhang, Z. (R&R). Neural Network Analysis of Psychological Data: A Step-by-Step Guide. Submitted to <i>Multivariate Behavioral Research</i> . [preprint] Tong, L. , & Zhang, Z. (preprint). TextSEM: Structural Equation Modeling for Text Data. Submitted to <i>Structural Equation Modeling: A Multidisciplinary Journal</i> . [R package] Tong, L. (preprint). Nonlinear Structural Equation Modeling with Text Data. <i>Dissertation</i> . <i>Published / In Press</i> Wan R.*, Tong, L. * (co-first author), Kneareem T., Li T., Huang K., & Wu Q. Hashtag Re-appropriation For Audience Control on Recommendation-driven Social Media Xiaohongshu. <i>The ACM Conference on Human Factors in Computing Systems (CHI)</i> . Honorable Mention for Best Paper (Top 5%). [paper] Tong, L. , Qu, W., & Zhang, Z. (2024). Comparison of the K1 Rule, Parallel Analysis, and the Bass-Ackward Method on Identifying the Number of Factors in Factor Analysis. <i>Fudan Journal of the Humanities and Social Sciences</i> , 1-28. [paper] Lu, Y., Tong, L. , & Cheng, Y. (2024). Advanced Knowledge Tracing for Intelligent Tutoring Systems: Incorporating Process Data and Curricula Information via an Attention-Based Framework for Accuracy and Interpretability. <i>Journal of Educational Data Mining</i> . [paper] Tong, L. , Liu, Q., Yu, W., Yu, M., Zhang, Z., & Jiang, M. (2023). Improving mental health support response generation with event-based knowledge graph. <i>Workshop on Knowledge-Augmented Methods for NLP (KnowledgeNLP) at AAAI Conference on Artificial Intelligence (AAAI)</i> . [paper] Jiang M., Dang H., & Tong L. (2023). A Quantitative Review on Language Model Efficiency Research. Large Language Model. <i>Symposium at International Joint Conference on Artificial Intelligence (IJCAI)</i> . [paper]	

Wan, R., & **Tong, L.** (2023). Digital and Historical Exclusivity in Feminine Linguistics: From Nùshu to Xiaohongshu. *WiNLP workshop at Conference on Empirical Methods in Natural Language Processing (EMNLP)*. [[paper](#)]

Yu, M., Yu, W., **Tong, L.**, & Jiang, M. (2022). Scientific Comparative Argument Generation. *Document Intelligence Workshop (DI) at ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. [[paper](#)]

Kuebler, J., **Tong, L.**, & Jiang, M. (2021). Multi-Round Parsing-Based Multiword Rules for Scientific Knowledge Extraction. *IEEE International Conference on Big Knowledge (ICBK)* (pp. 331-338). IEEE. [[paper](#)]

Chen, H., Jin S., Lin W., Zhu Z., **Tong L.**, Liu Y., ... & Sun M. (2021). Quantitative analysis on the Communication of COVID-19 Related Social Media Rumors. (In Chinese). *Journal of Computer Research and Development*, 58(7), 1366-1384. [[paper](#)]

PRESENTATIONS AND POSTERS

Tong, L. & Jiang, M. (2024). Large Language Models in Mental Health Support. Invited talk for *Summer Research Experience for Teacher's (RET) Program*, 19 June, Notre Dame, USA.

Tong, L., Zhang, Z., Jiang M., & Li J. (2023). Permutation test of Importance-Weighted Autoencoder for Factor Analysis. *Symposium at Lucy Family Institute for Data & Society*, 7 September, Notre Dame, USA.

Tong, L. & Zhang, Z. (2023). Evaluation of the Bass-Ackward Method for Identifying the Number of Factors. *Annual Meeting of the International Society for Data Science and Analytics*, 4–6 July, Shanghai, China.

Tong, L., Liu, Q., Yu, W., Yu, M., Zhang, Z., & Jiang, M. (2023). MHKG: Improving mental health support response generation with event-based knowledge graph. *KnowledgeNLP at AAAI*, 7–14 February, Washington DC, USA.

Zhang, J., Chen, H., & **Tong, L.** (2022). Structured Gender Bias: How is Gender Cognition Changed and Affected by Digital Communities in China? *Annual Conference of International Association for Media and Communication Research Conference (IAMCR)*, 11–15 July, Beijing, China (online).

Tong, L., Zhang, Z., Jiang M., & Li J. (2022). Estimating Structural Equation Models with Neural Networks. *Symposium at Lucy Family Institute for Data & Society*, 28 September, Notre Dame, USA.

SOFTWARE DEVELOPMENT

Tong L., & Zhang, Z. (2024). TextSEM: An R Package for Structural Equation Modeling with Text Data. [[code](#)]

Tong, L., Nguyen, B., Dang H., Hoq, A., & Taki S. BLASH: An LLM-based Chrome Plugin for Peer-to-Peer Mental Health Support in Online Forums. [[video demo](#)] [[code](#)]

Tong L., & Zhang, Z. (2022). An Online Tool for Bass-Ackward Factor Analysis. [[online app](#)]

Xu, J., **Tong L.**, & Zhang, Z. (2020). Webnetvis: An Online Tool for Network Visualization. [[online app](#)]

HONORS AND AWARDS

Center for Research Computing (CRC) Graduate Award (\$1,000) 2025
University of Notre Dame

ISLA Graduate Student Research Award (\$4,000) 2024

	University of Notre Dame	
	First place, 2023 EDM Cup	2023
	International Educational Data Mining Society (IEDMS)	
	Summer Institute of Computational Social Science (SICSS) Scholarship	2023
	NYU Shanghai Center for Applied Social and Economic Research (CASER)	
	Most Innovative Project Award	2019
	Google AI ML Winter Camp, Beijing	
	First Prize Academic Scholarship	2019
	Sichuan University	
	Outstanding Undergraduate Student Award	2017, 2018, 2019
	Sichuan University	
	National Innovation Training Grant for Undergraduates	2017, 2018
	Sichuan University	
TEACHING EXPERIENCE	Co-Instructor , Grad Seminar: Quantitative Study (PSY 63199) Department of Psychology, University of Notre Dame	Spring 2025, Fall 2024
	Teaching Assistant/Lab Instructor , Methods for Behavioral Sciences (PSY 30160) Department of Psychology, University of Notre Dame	Fall 2023, Spring 2022
MENTORED STUDENTS	- Noah Crockett, ND undergraduate student in psychology - Anna Krush, ND master's student in applied and computational mathematics and statistics - Yue Wan, ND exchange undergraduate student in data science - Ishita Masetty, Penn High School student - Qi Liu, ND visiting undergraduate student in computer science - Longqing Chen, ND undergraduate student in computer science	2024 2023 – 2024 2023 – 2024 2022 2022 2021
PROFESSIONAL LEADERSHIP & SERVICE	Workshop Co-Organizer: - The Second Workshop on Knowledge Augmented Methods for NLP at ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 6–10 August, 2023, Long Beach, CA, USA. [website] - Workshops at the Annual Meeting of the International Society for Data Science and Analytics, 21–24 July, 2024, Vienna, Austria. [website] Manuscript Reviewer: - Journal of Behavioral Data Science (JBDS) - Journal of Chinese Political Science (JCPS) - Annual Meeting of the Association for Computational Linguistics (ACL) - Conference on Empirical Methods in Natural Language Processing (EMNLP) - China National Conference on Computational Linguistics (CCL) University Service: - Departmental DEI Committee Representative - Mentor for STEMentorship Program, Association with Women in Science - Departmental Faculty Meeting Representative	2023 – 2024 2023 – 2024 2022 – 2023
TECHNICAL SKILLS	Programming Languages: Python, R, JavaScript, SQL, HTML and others	

- Quantitative Methods: Structural Equation Modeling, Multivariate Analysis, Longitudinal Data Analysis, Bayesian Statistics, Item Response Theory
- Machine Learning Topics: Explainable Machine Learning, Generative models, Dialog System, Empathetic AI, Human-AI Collaboration

REFERENCES

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